

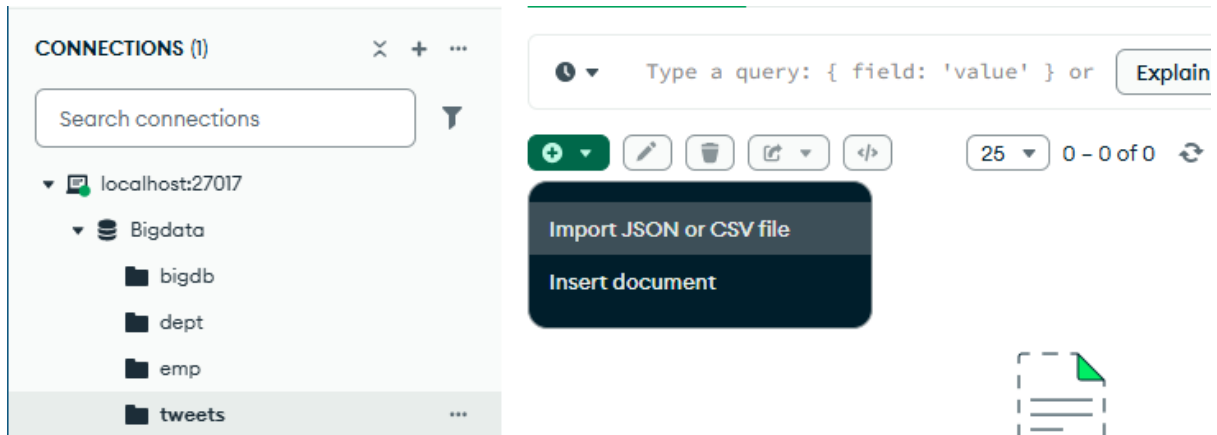
Module: Big Data

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1. Start MongoDB



```
> show dbs
< Bigdata  380.00 KiB
  admin    40.00 KiB
  config   72.00 KiB
  local    144.00 KiB
  saan_db  920.00 KiB
> use Bigdata
< switched to db Bigdata
> show collections
< bigdb
  dept
  emp
  tweets
> db.tweets.countDocuments()
< 250
Bigdata> |
```

Explore the Documents

```
> db.tweets.findOne()
< {
  _id: ObjectId('69c4bba016a5623b42842477'),
  contributors: null,
  truncated: false,
  text: 'Best January bus duty weather EVER! #lionnation #dasdpride https://t.co/cTyq7ib3Z6 https://t.co/5Bzh49ffAk',
  is_quote_status: false,
  in_reply_to_status_id: null,
  id: 1224424872627122200,
  favorite_count: 0,
  entities: {
    symbols: [],
    user_mentions: [],
    hashtags: [
      {
        indices: [
          36,
          47
        ],
        text: 'lionnation'
      },
      {
        indices: [
          48,
          58
        ],
        text: 'dasdpride'
      }
    ],
    urls: [
      {
        url: 'https://t.co/cTyq7ib3Z6',
        indices: [
          59,
          82
        ],
        expanded_url: 'https://ift.tt/3b9Stsk',
        display_url: 'ift.tt/3b9Stsk'
      }
    ]
  }
}
```

```

> db.tweets.distinct("lang")
< [ 'en', 'pt' ]
> db.tweets.distinct("user.name")
< [
  '#BernieorBust',
  '3DPat',
  '6',
  'AJM',
  'Abigailadams 🍷 ',
  'Aimee DeMaio',
  'Alec Luhn',
  'Alex 🌧️🌸🌧️ AU',
  'Alexis Moberger',
  'Alliandre.xl8',
  'Aloisia Anja',
  'An Edwardian Evans',
  'Andhi Pratama Putra',
  'Andre' D. Foxx",
  'Ann AU 🍷',
  'Aramayis Iritsyan',
  'Aunya 🇧🇷🇮🇸 (88%) ➡️ BTSM 2/14',
  'Babyyci 🦋',
  'Barrel Bitch',
  'Ben #Daisy Gang',
  'Bilzin Sumberg',
  'Brandon Essig',
  'Bri',
  'Brookie 🍷',
  'COAvsGirl07',
  'Cam Gillentine',
  'CanGal 🍷',
  'Captain Caveman',
  'Cathy Williamson',
  'Cha(rizar)d Black',
  'Chereen',
  'Cheryl Page',
  'Chris Turner',
  'Christine Padrid',

```

Search for specific field value

```
> db.tweets.find({"user.name":"puff"})
< {
  _id: ObjectId('69c4bba016a5623b428424bf'),
  contributors: null,
  truncated: false,
  text: 'psa: if you ever come out then dont forget to come back in cus the weather be gettin cold!',
  is_quote_status: false,
  in_reply_to_status_id: null,
  id: 1224389340664684500,
  favorite_count: 0,
  entities: {
    symbols: [],
    user_mentions: [],
    hashtags: [],
    urls: []
  },
  retweeted: false,
  coordinates: null,
  source: '<a href="http://twitter.com/download/iphone" rel="nofollow">Twitter for iPhone</a>',
  in_reply_to_screen_name: null,
  in_reply_to_user_id: null,
  retweet_count: 0,
  id_str: '1224389340664684544',
  favorited: false,
  user: {
    follow_request_sent: false,
    has_extended_profile: true,
    profile_use_background_image: false,
    default_profile_image: false,
    id: 1088571664407900200,
    profile_background_image_url_https: 'https://abs.twimg.com/images/themes/theme1/bg.png',
    verified: false,
    translator_type: 'none',
    profile_text_color: '000000',
    profile_image_url_https: 'https://pbs.twimg.com/profile_images/1210888484283662337/ssvQSHy5_normal.jpg',
    profile_sidebar_fill_color: '000000',
    entities: {
      url: {
        url: [
```

Filter fields returned by query

```
> db.tweets.find(
  {"user.name":"#BernieorBust"},
  {id:1, _id:0}
).pretty()
< {
  id: 1224302343509942300
}
```

Pattern Matching

```
> db.tweets.find({text:/sun/}).count()
< 4
> db.tweets.find({text:/rain/}).count()
< 13
Bigdata >
```

\$regex

```
> db.tweets.find({ text: { $regex: /sun/ } }).count()
< 4
Bigdata > |
```

Regular Expression Options

Start with "RT"

End with "snow"

```
> db.tweets.find({ text: { $regex: /sun/i } }).count()
< 9
> db.tweets.find({ text: { $regex: /^RT/i } }).count()
< 86
> db.tweets.find({ text: { $regex: /snow$/i } })
<
Bigdata >
```

2. Indexes

Creating an Index

```
> db.tweets.createIndex({text:"text"})
< text_text
> db.tweets.find({$text:{$search:"snow"}}).count()
< 6
Bigdata>
```

Searching with an Index

```
> db.tweets.find({
  "retweeted_status.retweet_count":{$gt:20}
}).count()
< 33
Bigdata>
```

Show Indexes

```
> db.tweets.getIndexes()
< [
  { v: 2, key: { _id: 1 }, name: '_id_' },
  {
    v: 2,
    key: { _fts: 'text', _ftsx: 1 },
    name: 'text_text',
    weights: { text: 1 },
    default_language: 'english',
    language_override: 'language',
    textIndexVersion: 3
  }
]
Bigdata> |
```

Dropping an Index

```
> db.tweets.dropIndex("text_text")
< { nIndexesWas: 2, ok: 1 }
Bigdata>
```

3. Further Queries

Operators

```
> db.tweets.find({"retweeted_status.retweet_count": {$gt:20}}).count()
< 33
Bigdata > |
```

Null & Boolean

```
> db.tweets.find({"user.location":{$eq:null}}).count()
< 0
> db.tweets.find({"user.geo_enabled":true}).count()
< 117
Bigdata >
```

Where Clause

```
> db.tweets.find({$where : "this.user.followers_count > this.user.friends_count" }).count()
< 115
Bigdata > |
```

Nested documents

```
> db.tweets.find({
  "retweeted_status.user.followers_count":{$gt:2000}
}).count()
< 64
Bigdata > |
```


4. Group by Commands and Aggregation

```
> db.tweets.aggregate([
  {
    $group:{
      _id:null,
      total:{$sum:"$retweeted_status.retweet_count"},
      count:{$sum:1}
    }
  }
])
< {
  _id: null,
  total: 2210,
  count: 250
}
Bigdata>
```

5. Error Checking

```
> db.tweets.find({$and : [ {text : /ice/}, { "retweeted_status.retweet_count": { $gt: 50}}]}).count()
❌ > SyntaxError: Unexpected token, expected "," (1:88)

> 1 | db.tweets.find({$and : [ {text : /ice/}, { "retweeted_status.retweet_count": { $gt: 50}}]}).count()
    |                                                                                                     ^
    2 |
> db.tweets.find({
  text: /ice/,
  "retweeted_status.retweet_count": { $gt: 50 }
}).count()
< 0
Bigdata>
```

6. Cleaning Data

```
> db.tweets.updateMany(
  {},
  [
    { $set: { country: { $ifNull: ["$place.country", "Unknown"] } } }
  ]
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 250,
  modifiedCount: 250,
  upsertedCount: 0
}
Bigdata > |
```

Updating One Document

```
> db.tweets.updateOne(
  { _id: ObjectId("69c4bba016a5623b42842477") },
  { $set: { lang: "English" } }
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
Bigdata > |
```

Undefined Language

```
> db.tweets.updateMany(
  {lang:"und"},
  {$set:{lang:"unknown"}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 0,
  modifiedCount: 0,
  upsertedCount: 0
}
Bigdata >
```

Nested Documents

```
> db.tweets.updateMany(
  {'user.location':{$regex:'England',$options:'i'}},
  {$set:{'user.location':"UK"}}
)
< {
  acknowledged: true,
  insertedId: null,
  matchedCount: 2,
  modifiedCount: 2,
  upsertedCount: 0
}
Bigdata >
```